

KUMAR RAJ

Software Engineer | Backend Developer | Full Stack Developer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [GeeksforGeeks](#)

PROFESSIONAL SUMMARY

Backend-focused Software Engineer experienced in designing and deploying full-stack, production-ready applications using FastAPI, Python, and MySQL. Skilled in building secure REST APIs, real-time features with WebSockets, and scalable relational database architectures. Proven ability to take SaaS products from system design through production deployment, with a strong foundation in authentication, authorization, and API security. Passionate about solving real-world problems through clean, maintainable, and scalable engineering solutions.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML, CSS,

Backend & APIs: FastAPI, REST API Design, JWT Authentication, SQLAlchemy ORM, WebSockets, Uvicorn, Rate Limiting

Frontend & Database: HTML, CSS, JavaScript, Responsive Design, MySQL, Database Design

Core Concepts: Data Structures, OOP, Authentication & Authorization, API Development, SEO, Deployment

Libraries & Tools: NumPy, Pandas, Matplotlib, Git, GitHub, Render

PROJECTS

Apna Bhagalpur — Clinic & Hospital Appointment Management SaaS | [Live Demo](#)

- Architected and deployed a full-stack SaaS platform for clinics and hospitals, enabling digital appointment management across three role-based portals: Admin, Clinic, and Patient dashboards.
- Designed secure JWT-based authentication and authorization with password hashing to protect multi-role access across the platform.
- Built a real-time appointment update system using WebSockets, syncing status changes instantly between clinic and patient dashboards without page refresh.
- Engineered clean REST APIs with FastAPI and modeled relational data using SQLAlchemy ORM over MySQL, maintaining clear separation between API, business logic, and data layers.
- Implemented rate limiting to guard against API abuse and shipped a responsive UI to production.

Bhagalpur Stays — Hotel Booking Platform | [Live Demo](#)

- Built a full-stack hotel booking platform with property listings, an owner dashboard, an admin dashboard, and end-to-end reservation workflows.
- Designed booking, wishlist, and coupon systems on FastAPI and MySQL to handle reservation logic and promotional pricing.
- Developed a review and rating system to support user-generated feedback and improve booking decisions.
- Implemented role-based secure authentication for distinct owner, admin, and guest workflows, and deployed a responsive, dynamic-content frontend to production.

Yatra Planner — Intelligent Travel Budget Platform | [Live Demo](#)

- Built a frontend-driven travel budget planning platform from scratch using HTML, CSS, and JavaScript, with no backend framework.
- Designed a dynamic budget calculation engine supporting multiple departure cities and real-time cost comparisons.
- Integrated live weather data and a travel guide/directory to support itinerary planning with real-world context.
- Implemented a premium PDF generation feature for exporting personalized travel plans.
- Optimized the platform for SEO and performance, and deployed a fully responsive UI to production.

EXPERIENCE

HCL Technologies — Training Program December 2024 – January 2025

- Completed a structured technical training program covering industry software development practices, tools, and workflows.
- Strengthened foundational engineering practices applicable to backend development and production software delivery.

EDUCATION

KCC Institute of Technology & Management

Bachelor of Technology, Computer Science Engineering (AI & Machine Learning) | 2022 – 2026 | CGPA: 7.39/10

CERTIFICATIONS

- HCL Technologies — Training Program Certificate

CODING PROFILES

[GitHub](#) | [GeeksforGeeks](#) | [Portfolio](#)